

Kevin Maresca

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Versatile, self-starter engineer with deep understanding of multiple US energy markets, with strong problem-solving, data-analysis, and applied systems design skills.

PROFESSIONAL EXPERIENCE

Engineer III (multiple)

PEPCO (Exelon), Washington, DC

Feb. 2023 - April 2026

Reliability Engineer

- Lead daily reviews of complex distribution outages, performing root-cause failure analysis, data validation, and customer impact mitigation.
- Identified trends and addressed recurring issues through data analysis, ensuring corrective actions for emergent engineering challenges.
- Managed emergent work-plans for critical customer complaints, storms, hazardous conditions, and high-priority government events.
- Independent Projects:
 - *WattSure* - a full-stack web-application used to automate and standardize outage data workflows and reporting
 - *"A Case for the System Health Index"* - technical whitepaper defining a KPI to track health of distribution equipment. The algorithm was adopted by the Asset Health Scoring Initiative.

Lead Responsible Engineer

- Oversaw multi-million-dollar system improvement programs, managing prioritization and execution of engineering design deliverables.
- Reviewed and approved engineering job-jackets, ensuring alignment with project requirements, technical standards, and constructibility constraints.

Distribution Designer

ENERCON Services, Kennesaw, GA

Oct. 2021 - Dec. 2022

- Executed electrical distribution design work under contract for six U.S. utilities.
- Executed distribution designs from ideation through closeout, including field surveys, database reconciliation (ROWS, property lines), and preparation of construction-ready deliverables.
- Designed sustainable, efficient, cost-effective solutions to meet customer requests. Job-jacket deliverables included construction prints, inventories, easement contracts, permits, and cost estimates.

Research Assistant

University of New Mexico, Albuquerque, NM

Jan. 2021 - June 2022

- Researched high-efficiency self-driving vehicle algorithms to minimize computation-time for decision-making.
 - Studied use of Lyapunov and invariant sets to define driving environment, simulated in MATLAB, achieving a functional algorithm for parking and highway conditions.
 - Validated algorithm on small-scale experiments with RC vehicles in controlled environments.

Engineering Summer/Winter Intern

Los Alamos National Laboratory, Los Alamos, NM

May 2016 - Aug. 2020

- Designed and constructed nuclear material aging containment vessels, weapons components, and shipping containers for transport of shockwave-induced spallation test unit experiments.
- Performed finite element analyses to validate performance across full product lifecycle

EDUCATION

B.S. Mechanical Engineering (cum laude) with Minor in Mathematics

University of New Mexico, Albuquerque, NM

Aug 2017 - May 2021

SKILLS

Languages: English (native), Italian (native), Spanish

CAD Design: SolidWorks, PTC Creo, AutoCAD, Adobe Acrobat, Bluebeam, Model-Based Definition principles

Finite Element Analysis - Creo Simulate, ANSYS (CFD & FEA), Cubit

Electric Distribution Design/Analysis - GWD, ArcGIS, G-Technology, DistView, PoleForeman, ADMS, OMS, iTron

Data Analysis - Excel, MATLAB, Power Bi, Javascript, NodeJS